

3	(a) obeys the equation $pV/T = \text{constant}$ (accept $pV = nRT$)	B1	[1]
(b)	(i) $pV = nRT$ $5.0 \times 10^7 \times 3.0 \times 10^{-4} = n \times 8.31 \times 296$ giving $n = 6.1 \text{ mol}$	C1	[2]
	(ii) pressure $\propto$ amount of substance loss = $0.40/100 \times 6.1 \text{ mol} = 0.0244 \text{ mol}$ = $0.0244 \times 6.02 \times 10^{23}$ (atoms) = 1.47×10^{22} atoms	C1	
	rate = $(1.47 \times 10^{22}) / (35 \times 24 \times 60 \times 60)$ = $4.9 \times 10^{15} \text{ s}^{-1}$	A1	[4]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2014	9702	41